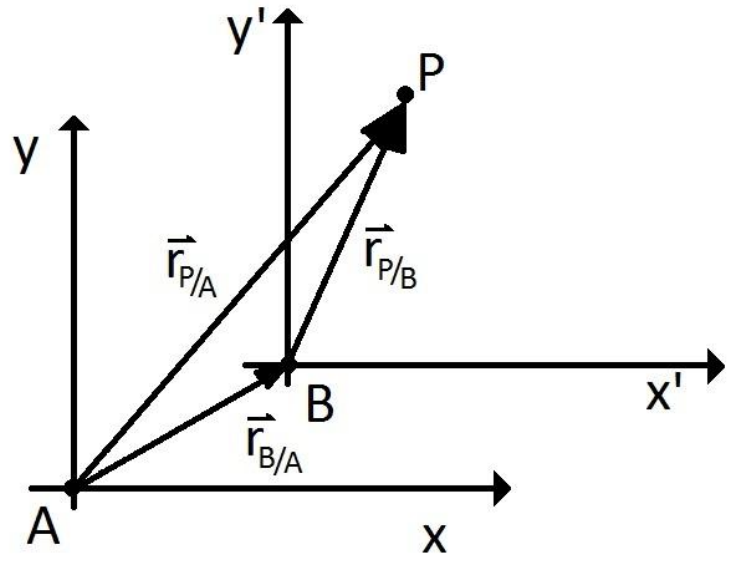
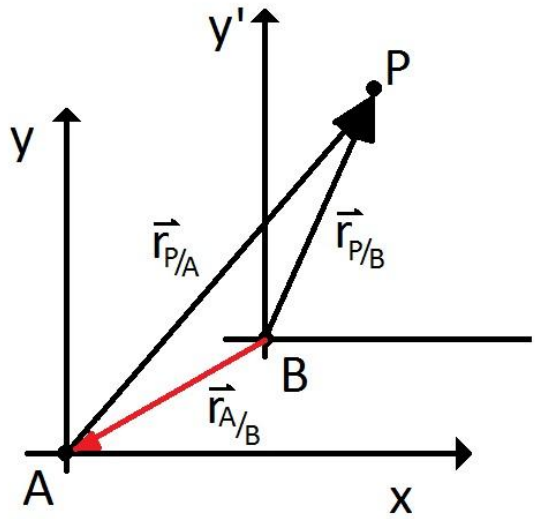


$$\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{r}_{B/A} \quad (1)$$

$$\frac{d\vec{r}_{P/A}}{dt} = \frac{d\vec{r}_{P/B}}{dt} + \frac{d\vec{r}_{B/A}}{dt}$$

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A} \quad (2)$$

$$\vec{a}_{P/A} = \vec{a}_{P/B} + \vec{a}_{B/A} \quad (3)$$



$$\vec{r}_{P/B} = \vec{r}_{P/A} + \vec{r}_{A/B} \quad (4)$$

$$\vec{v}_{P/B} = \vec{v}_{P/A} + \vec{v}_{A/B} \quad (5)$$

$$\vec{a}_{P/B} = \vec{a}_{P/A} + \vec{a}_{A/B} \quad (6)$$

$$\vec{r}_{P/A} = \vec{r}_{P/B} + \vec{r}_{B/A} \quad (1) \qquad \vec{r}_{P/B} = \vec{r}_{P/A} + \vec{r}_{A/B} \quad (4)$$

$$\cancel{\vec{r}_{P/A}} = \cancel{\vec{r}_{P/A}} + \vec{r}_{A/B} + \vec{r}_{B/A}$$

$$0 = \vec{r}_{A/B} + \vec{r}_{B/A}$$

$$\boxed{\vec{r}_{A/B} = -\vec{r}_{B/A}}$$

$$\vec{v}_{A/B} = -\vec{v}_{B/A}$$

$$\vec{a}_{A/B} = -\vec{a}_{B/A}$$